Proposal

Enhancing the Safety and Authenticity of Online Discussions on Social Media Platforms with AI

Introduction

Social media platforms have revolutionized communication and information sharing, making them central to modern life. However, they are increasingly used to spread fake news and facilitate online threats, posing significant challenges to societal trust, safety, and the quality. Fake news misleads the public and manipulates opinions, while online threats, such as hate speech and cyberbullying, foster hostility and harm. These issues often overlap, amplifying their impact and making their detection a critical priority.

Current detection systems primarily focus on fake news and threats as separate problems. While this approach has led to advancements in combating these issues individually, it overlooks the interconnected nature of misinformation and harmful behavior. Addressing these challenges together is essential to develop more comprehensive solutions.

Transformer-based models, particularly BERT, have shown remarkable success in text classification tasks. However, their application to multitask detection of fake news and threats remains underexplored. This research seeks to fill this gap by evaluating the performance of separate versus joint detection models and analyzing the trends in transformer architectures, such as encoders, decoders, and hybrids.

By enhancing detection methods, this study aims to contribute to safer online environments, and foster informed public engagement. The findings will provide valuable insights into AI-driven solutions for improving social media safety and the integrity of digital spaces.

Problem Statement

The spread of misinformation and harmful behaviors on social media platforms has created significant challenges to fostering safe and authentic online environments. The rapid dissemination of false information fake news manipulates public opinion, undermines societal trust, and exacerbates the polarization of communities. Meanwhile, threatening and abusive interactions erode the quality of discourse, deterring meaningful engagement and amplifying negativity.

Fake news and online threats are often studied as separate problems, but in reality, they can overlap. For example, fake news can incite harmful behavior, and threats can spread false information. Existing approaches fail to address this overlap effectively. Moreover, different types of transformer models, such as encoders, decoders, and hybrid architectures, have not been compared for this task. This study will address these gaps by exploring the best approaches for combined detection and analyzing their impact on online safety & security.

Research Questions

Q1. How does the performance of separate BERT models trained individually on fake news and threat datasets compare to a single BERT model trained jointly on both datasets and which approach yields better results for combined detection tasks?

Q2. How can the conclusions be drawn from the combined detection of fake news and threats contribute to improving societal safety, fostering informed public engagement, and enhancing the quality of social media?

Q3. What are the trends in different transformer models (encoder, decoder, hybrid) while detecting online threats & fake news detection?

Literature Review

The challenges posed by misinformation and online threats have been extensively studied, but their overlap and combined detection remain underexplored. This section reviews recent works on fake news detection, threat detection, and the application of transformer-based models, connecting them to the research questions outlined.

Fake News Detection: Recent work on disinformation, such as the *DeFaktS* dataset, highlights the importance of fine-grained detection using transformer models like BERT. This study is relevant to Research Question 1, as it informs the comparison of separate versus joint BERT models for fake news detection. [1]

Threat Detection: The survey on hate speech and toxicity detection focuses on the growing role of transformer models in identifying harmful content. It provides insights into encoder-based and hybrid architectures, relevant to Research Question 3, exploring trends in transformer models for threat detection. [2]

Social Media Safety: The use of semantic networks in threat modeling, discussed in *Threat Modelling and Detection Using Semantic Network for Improving Social Media Safety*, enhances the understanding of harmful online behavior. Although it doesn't address combined detection, its methods inform the development of multitask detection systems, aligning with Research Question 1. [3]

This literature highlights the potential of transformer models, particularly BERT and hybrid architectures, to enhance detection accuracy and efficiency in both areas. This research aims to fill that gap by exploring the comparative effectiveness of separate versus joint models, analyzing the trends in transformer architectures, and the broader societal benefits of improving digital safety.

Objectives

1. **Evaluate the Effectiveness of Multi-task Learning**: Assess the comparative performance of separate versus unified BERT models in detecting fake news and online threats. This involves training:

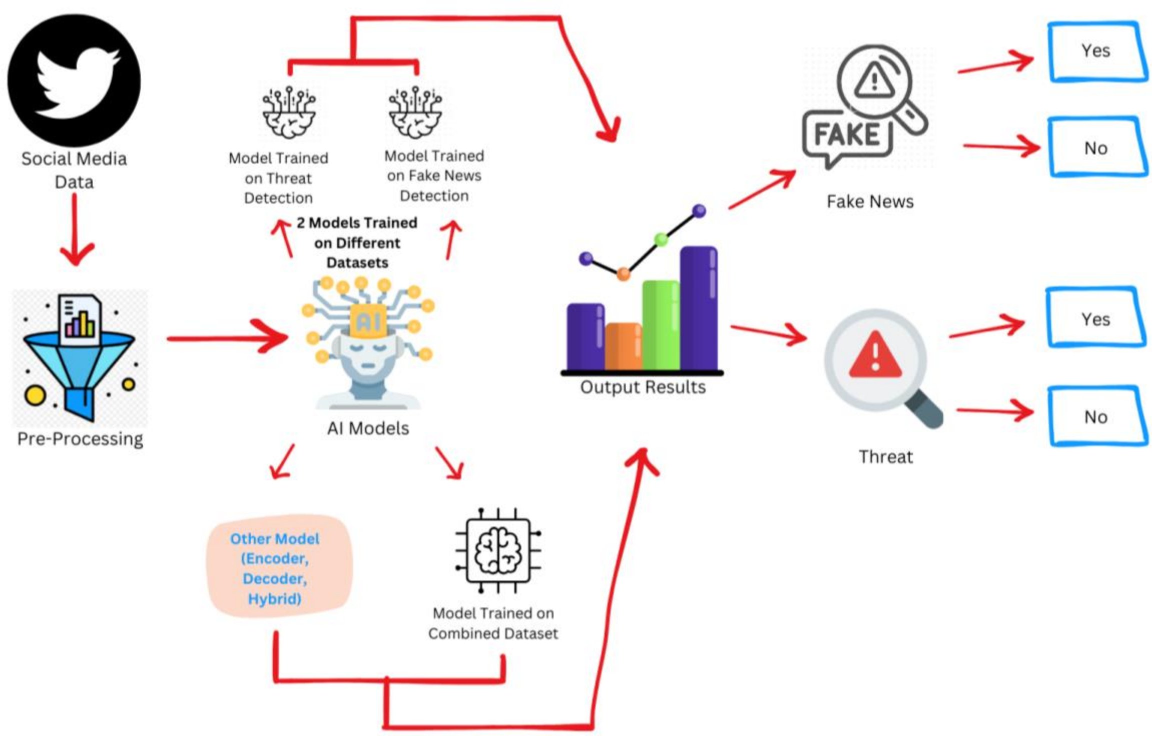
    - Dedicated models specialized for individual tasks (fake news and threats).
    - A multitask BERT model addressing both tasks simultaneously.

2. **Model Optimization for Combined Detection**: Investigate how integrating fake news and threat detection within a single framework enhances model efficiency, improves predictive accuracy, and minimizes computational overhead.

3. **Impact on Online Safety and Awareness**: Measure the societal impact of improved detection mechanisms in reducing harmful content, raising public awareness, and fostering safer social media interactions.

4. **Explore Transformer Architectures**: Analyze the capabilities and trends of advanced transformer models (encoder-based, decoder-based, and hybrid architectures) in handling the complexity of misinformation and harmful content.

Diagram 1:



Scope of Metrics:

The effectiveness of the models will be evaluated based on:

1. **Fake News Detection Metrics:** To evaluate fake news detection, the system will analyze textual features derived from disinformation indicators, including but not limited to.

    - **Semantic Inconsistency:** Identifying contradictions or logical errors within the content to flag unreliable posts.
    - **Polarization:** Detecting friend-foe framing narratives in fake news articles.
    - **Vagueness of Phrasing:** Analyzing the use of indirect expressions or hedging words.
    - **Authenticity and Referencing of Information:** Measuring the use of reliable references versus false or absent contextualized sources.
      These features represent key characteristics for distinguishing fake news content, ensuring that the system can provide good detection capabilities.

2. **Threat Detection Metrics:** The threat detection component focuses on identifying harmful online behaviors and speech. The system will be evaluated based on.

    - **Toxicity:** Identifying harmful and aggressive language.
    - **Threat:** Detecting language that explicitly or implicitly poses harm to individuals or groups.
    - **Insult:** Recognizing disrespectful or degrading comments.
    - **Identity Hate:** Identifying hate speech targeted toward individuals or groups based on identity attributes like race, religion, or gender.
      These metrics ensure that the model addresses both explicit and implicit threats within the online discourse, enhancing safety.

3. **Content Classification Accuracy:**
   The model's ability to classify different types of fake news (e.g., misinformation vs. disinformation) and threats (e.g., hate speech vs. cyberbullying).
4. **Security:** The models' ability to improve online platform safety by reducing the spread of harmful content.
5. **Societal Safety:** The broader impact on public awareness, informed decision-making, and fostering safer online environments.

Methods:

1. **Input Data Collection and Preprocessing:**

    - **Data Sources**: Social media platforms dataset will be used as well as datasets related to fake news (e.g., DeFaktS) and harmful content (e.g., toxicity, hate speech, and threats) will be integrated.
    - **Preprocessing**:
        - Tokenization and cleaning of text to remove noise such as special characters, URLs, and stop words.
        - Annotation of data for binary outputs: Fake News (1 or 0) and Threats (1 or 0).
        - Linguistic and semantic feature extraction based on key indicators such as semantic inconsistency, polarization, emotionalization, and vagueness of phrasing.
2. **Model Training Approaches**:
    - **Separate Models**: Train individual BERT models for fake news and threat detection tasks. Each model will be fine-tuned on its respective dataset.
    - **Unified Multi-task Model**: Implement a multitask BERT model using shared encoder layers and task-specific for fake news and threat detection.
    - **Comparison Metrics**: Evaluate models using accuracy, F1-score, precision, recall, and accuracy to determine performance.

3. **Transformer Architecture Exploration**: Experiment with encoder-based (BERT), decoder-based (GPT), and hybrid models for comparative performance analysis.

4. **Output Generation**: The models will produce binary outputs for each task:
    - Fake News Detection: Fake (1) or Not Fake (0).
    - Threat Detection: Threat (1) or No Threat (0).

5. **Evaluation of Combined Detection Impact**:
    - Measure improvements in predictive performance when fake news and threats are detected simultaneously.
    - Conduct comparative analysis on datasets processed through separate versus unified models to assess accuracy gains and computational efficiency.

6. **Societal Analysis:** Study how these methods can help make online spaces safer and improve the way people interact on social media.

Conclusion

This research will help us understand the best way to detect both fake news and online threats at the same time. It will show whether separate or joint models work better and highlight the trends in transformer models for these tasks. The findings will also explore how better detection can make online spaces safer and encourage healthy public discussions.

References:

1. Baly, R., Horne, B. D., & Adali, S. (2023). *DeFaktS: A German Dataset for Fine-Grained Disinformation Detection through Social Media Framing.* In Proceedings of the 2023 International Conference on Data Science and Advanced Analytics (DSAA). IEEE. Link

2. Gupta, S., & Chauhan, V. (2022). *Hate Speech, Toxicity Detection in Online Social Media: A Recent Survey of State of the Art and Opportunities.* Social Media Studies, 12(3), 45-63. Link

3. Singh, R., & Sharma, K. (2023). *Threat Modelling and Detection Using Semantic Network for Improving Social Media Safety.* Journal of Cybersecurity Studies, 5(4), 198-210. Link